

Aakash Choudhary

[in Aakash](#) | [✉ aakashc.work@gmail.com](mailto:aakashc.work@gmail.com) | [☎ +91-9027219749](tel:+91-9027219749) | [✉ @aakashc.dev](mailto:@aakashc.dev) | [🐙 ifaakash](#)

SUMMARY

Results-driven SRE/DevOps Engineer with 4+ years in cloud infrastructure automation, CI/CD, and container orchestration across AWS and Azure. Proficient in Terraform, Docker, Kubernetes, and ECS for scalable, highly available deployments. Experienced in FinOps practices, AWS MAP program management, and data engineering with Azure Data Factory. Hands-on with LLMOps — self-hosted LLMs on edge hardware, multi-model routing, and inference observability powering GenAI-driven monitoring and debugging — and building a self-directed platform-engineering lab around JIT access automation, GitOps, and CI/CD orchestration.

WORK EXPERIENCE

DevOps Engineer, Particle41

Feb 2025 - Present

- Architected and deployed AWS ECS-based services for highly available infrastructure, ensuring seamless scaling and fault tolerance across production workloads for clients including Outcomes (pharmacy tech) and Optic Systems (property inspection tech).
- Managed large-scale database migrations (1 TB) using AWS DMS and Amazon MSK, handling both homogeneous and heterogeneous migrations with reliable streaming and minimal downtime.
- Designed cost-saving strategies for idle and low-traffic workloads, achieving up to 40% monthly cost reduction; led AWS MAP funding initiatives collaborating with the AWS Cost team to secure project financing.
- Created robust CI/CD pipelines with automated triggers and branching strategies, decreasing deployment time by 50% and enhancing release reliability.
- Integrated SonarCloud and CodeRabbit into the CI pipeline, enabling automated code quality analysis on pull requests and reducing review effort by 80%.
- Integrated GenAI/LLM tools into batch scripts to automate monitoring alerts, log analysis, and debugging workflows across environments.

SRE/DevOps Engineer, Sopra Steria India

July 2022 - Feb 2025

- Led migration of on-premises infrastructure to AWS and Azure, improving scalability and achieving 25% cost reduction through automated resource management.
- Built CI/CD pipelines with GitLab CI/CD, reducing deployment time by 35%; designed microservices architecture with Docker and EKS/AKS, improving availability by 50%.
- Established monitoring and observability framework using Prometheus, Grafana, and Datadog, improving incident response time by 25% through custom dashboards and alerting.
- Implemented multi-cloud IaC with Terraform, creating reusable modules that reduced manual intervention by 50% and ensured consistent deployments.

PROJECTS

Edge LLMOps on Raspberry Pi 5

- Self-hosted Qwen2.5 (1.5B / 3B) LLMs via Ollama on a Raspberry Pi 5, running full inference on-device at the edge with no external API dependency.
- Benchmarked model and hardware performance across model sizes, tracking key latency metrics including time-to-first-token (TTFT) and tokens-per-second (TPS) throughput.
- Instrumented the stack with Node Exporter, Prometheus, and Grafana for real-time observability of inference latency, CPU/memory usage, temperature, and power draw.
- Configured LiteLLM as a unified proxy to route requests across multiple models through a single OpenAI-compatible API, and built a FastAPI service exposing model inference endpoints for integration into downstream applications.

Quick Spin

- Built an EC2 launch pad that combines Terraform and Ansible, allowing users to define instance count, type (public/private), and software dependencies (Java, NGINX, Apache, OpenSSH) in a single YAML file — deploying fully configured instances with one click.
- Automated Terraform plan/apply/destroy through a GitHub Actions CI/CD pipeline backed by S3 remote state, with a Tailscale-bootstrapped, dynamic AWS EC2 inventory driving Ansible runs.
- Built an Ansible role for just-in-time (JIT) bastion SSH access — provisioning and revoking lab users on demand with input-validated (`assert`), idempotent, least-privilege task design.

LabOps Hub

- Designing a self-hosted platform-engineering lab that mirrors a production DevOps control plane: JIT access automation in place, building out AWX-based playbook orchestration, ArgoCD GitOps delivery, Jenkins CI/CD, and Helm-packaged workloads on k3s.

k3s Hybrid-Cloud Homelab

- Ran production-style k3s (lightweight Kubernetes) workloads across a hybrid topology, with a control-plane node on an AWS EC2 VM and a locally managed worker node.
- Established secure, zero-trust connectivity between nodes using Cloudflared (Cloudflare Tunnel) and Tailscale, eliminating the need to expose public ports.
- Self-hosted Forgejo (Git forge) alongside the cluster for hands-on DevOps experimentation.

AI Committer

- CLI tool that suggests Git commit messages using a locally-run LLM or API key, streamlining version control workflows for the DevOps team.

GitHub Pruner

- CLI tool that cleans up GitHub Actions cache, workflow lists, and stale runs to keep repository history tidy and reduce storage overhead.

TECHNICAL SKILLS

Cloud Platforms:	AWS (EC2, S3, VPC, IAM, RDS, ECS, Glue, Lambda, EKS, DMS, MSK, CloudWatch), Azure (VMs, AKS, Blob Storage, VNets, Azure AD)
Data Engineering:	Azure Data Factory, Azure Data Lake Storage, AWS DMS, Amazon MSK
Infrastructure as Code (IaC):	Terraform (Cloud resource configurations)
Containerization & Orchestration:	Docker, Kubernetes (EKS/ECS), k3s (hybrid-cloud clusters), Helm
CI/CD:	GitHub Actions, GitLab CI/CD, Jenkins, AWS CodePipeline
Configuration Management:	Ansible (idempotent role design, FQCN modules, input-validated playbooks, dynamic & static inventories, JIT access automation), AWX (Ansible Tower), ArgoCD (GitOps)
LLMOps & GenAI:	Ollama (self-hosted LLMs), LiteLLM (multi-model routing), FastAPI (model serving), TTFT & tokens/sec benchmarking, edge inference; LLM-powered automation in shell scripts, AI-assisted debugging
Monitoring & Observability:	Prometheus, Grafana, Node Exporter, Datadog, Coralogix, Sentry
Linux & Networking:	Linux administration, networking fundamentals, Cloudflared (Cloudflare Tunnel), Tailscale, port management, firewall configuration
Scripting/Programming:	Bash (Automation scripting), Python, Java, scikit-learn (ML basics)
Version Control:	Git, Forgejo (self-hosted)
Database Technologies:	SQL (Database management, querying)

EDUCATION

Bachelor’s Degree in Computer Science and Engineering (CSE), ABESIT	(2018 - 2022)
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CERTIFICATIONS

- HashiCorp Certified: Terraform Associate (003) [click here](#)
- Microsoft Certified: Azure Fundamentals AZ-900 [click here](#)
- AWS Certified Solutions Architect – Associate

ADDITIONAL ROLES & CONTRIBUTIONS

Technical Writer, Plain English	June 2025 – Present
• Writing and publishing technical articles for a Medium publication with over 3.5M+ readers, focusing on DevOps, cloud infrastructure, and automation.	

PROFESSIONAL INTERESTS

- FinOps practices and cloud cost optimization for production environments.
- Exploring GenAI/LLM integration to elevate DevOps automation and observability.
- Building open-source CLI tools that improve DevOps team workflows and efficiency.